ERIC HOU

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EDUCATION

PURDUE UNIVERSITY, West Lafayette, IN — *Present - Dec 2027 (Expected)*

- GPA: 3.94 / 4.00 | B.S. in EE, College of Engineering

MISSION SAN JOSE HIGH SCHOOL, Fremont, CA — *Aug 2021 - Jun 2025*

- GPA: 4.38 Weighted | Relevant Coursework: Electrical Engineering Fundamentals, Digital System Design, Differential Equations
- SAT Score: 1550

RESEARCH

First Author - IEEE Paper — *Sep 2023 - Dec 2024*

- Engineered a vertical wall-scaling quadrupedal robot with adaptive suction for infrastructure monitoring with 9-lb payload
- Published and presented findings as first author at the 2024 IEEE IRC Conference

Aspiring Scholars Directed Research Program - Student Researcher — *Aug 2021 - Jun 2023*

- Selected for prestigious research program with 15% acceptance rate in that term's cohort
- Applied Quantum Annealing techniques to isolate particle collision jets using regression models and neural nets w/ Dr. McMahan

PROJECTS

5-Stage Electromagnetic Coilgun — *Jan 2025 - May 2025*

- Engineered multi-stage coilgun using Fusion 360 and KiCAD for high-current supercapacitor discharge control PCB
- Optimized magnetic flux using MATLAB simulations and Fusion 360
- Implemented rotating base and synchronized triggering for optimized projectile velocity, achieving 1.7m range with a 1lb payload

Autodesk Fusion 360 Wire Generator Plugin — *Oct 2023 - Feb 2024*

- Developed python script using Fusion 360 API to generate parameterized spline paths and wiring layouts
- Published plugin on Autodesk Store → 3,000+ active users, 5-star rating

EXPERIENCES

Avionics Engineer — *Aug 2025 - Present*

Active Controls | Purdue Space Program (PSP)

- Designed power monitor & control board for ASTRA, a self-stabilizing and landing EDF aerial vehicle

Hardware/Electrical Engineer — *Aug 2025 - Present*

Purdue Embedded Systems | SlayterHIL Project

- Designing hardware-in-the-Loop (HIL) testing infrastructure for drone flight controllers, simulating sensors and motor signals
- Prototyping communications and power distribution PCBs to interface between DUT and test node hardware

FTC TEAM 13356 — *Jul 2021 - Jun 2025*

Hardware Co-captain

- Directed end-to-end hardware design, manufacturing, and testing for competitive robotics seasons
- **Achievements:** #3 in California, Top 30 worldwide; qualified for Maryland Tech Invitational (top 0.5%) three years running
- Prototyped, manufactured, and shipped 300+ custom STEM kits for global organizations
- Developed and taught CAD/programming curriculum to Afghan refugees, Taiwanese Schools, and rookie teams
- Organized logistics for 8+ community events, reaching over 2000+ people

LEARNIFY+ Nonprofit — *Sep 2021 - Jan 2024*

Head of Computer Science & Board of Director

- Led a team of 50+ tutors delivering classes to 200+ students across 10 states and 4 countries
- Created course curricula in Java, Python, and HTML/CSS
- Volunteered 650+ hours in curriculum design, organizational leadership, and marketing

OTHER

SASA (Student Association for Stem Advocacy) — *Feb 2023 - May 2025*

Logistics Manager / Lobbyist

- Coordinated advocacy campaigns and conferences; lobbied senators and congressional offices for STEM education funding

HM Award - Toshiba ExploraVision Competition

- Developed design and business proposal for eco-friendly air purifying system using algae photobioreactor modules

MSJ Robotics Club President

- Planned and hosted club meetings; coordinated outreach events and fundraisers; interviewed candidates

SKILLS

Design / Manufacturing: KiCAD, FDM-3D Printing, Waterjet, CNC, Fusion 360 CAD/CAM/FEA

Programming Languages: Python, Java, C, HTML/JS/CSS